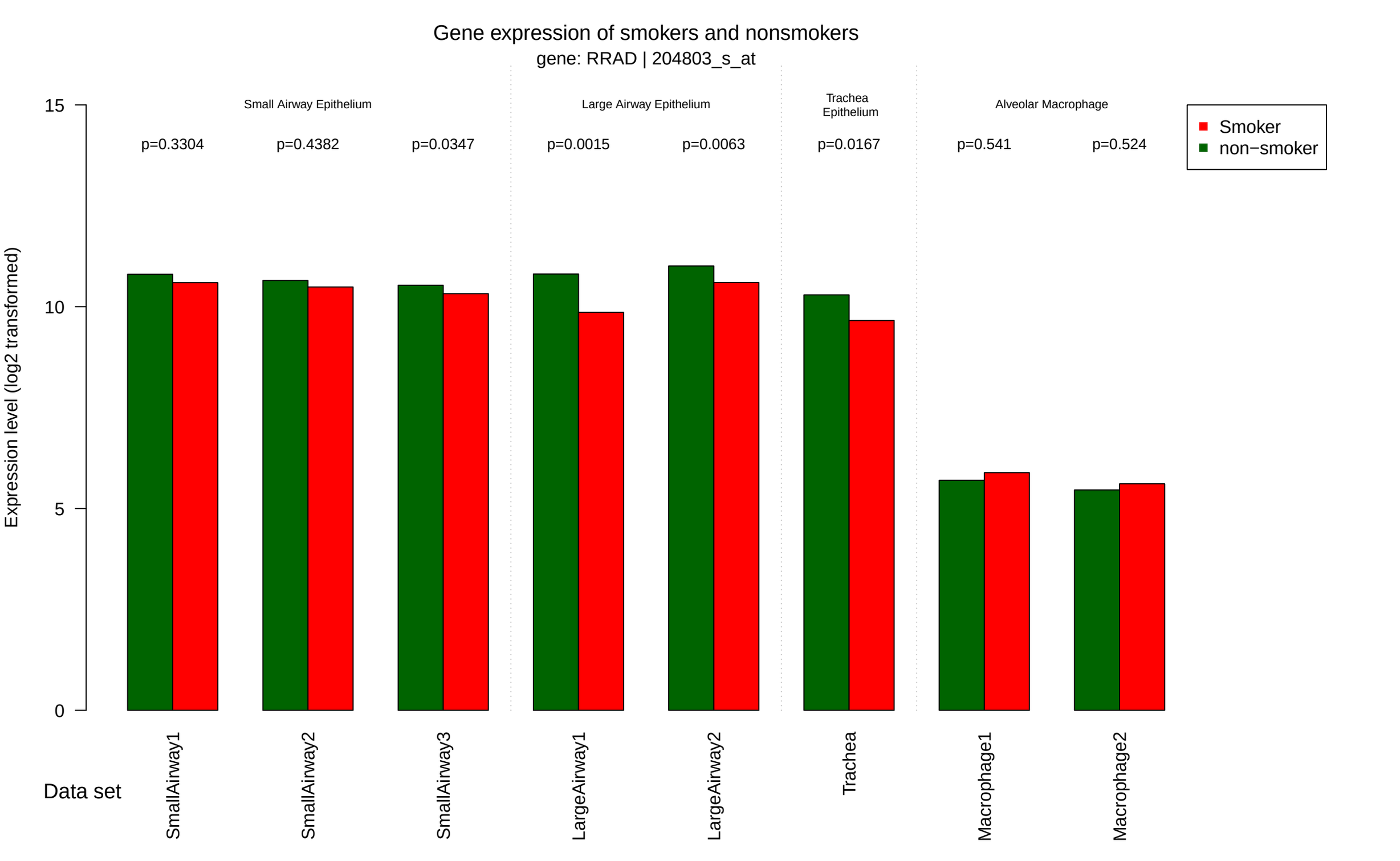
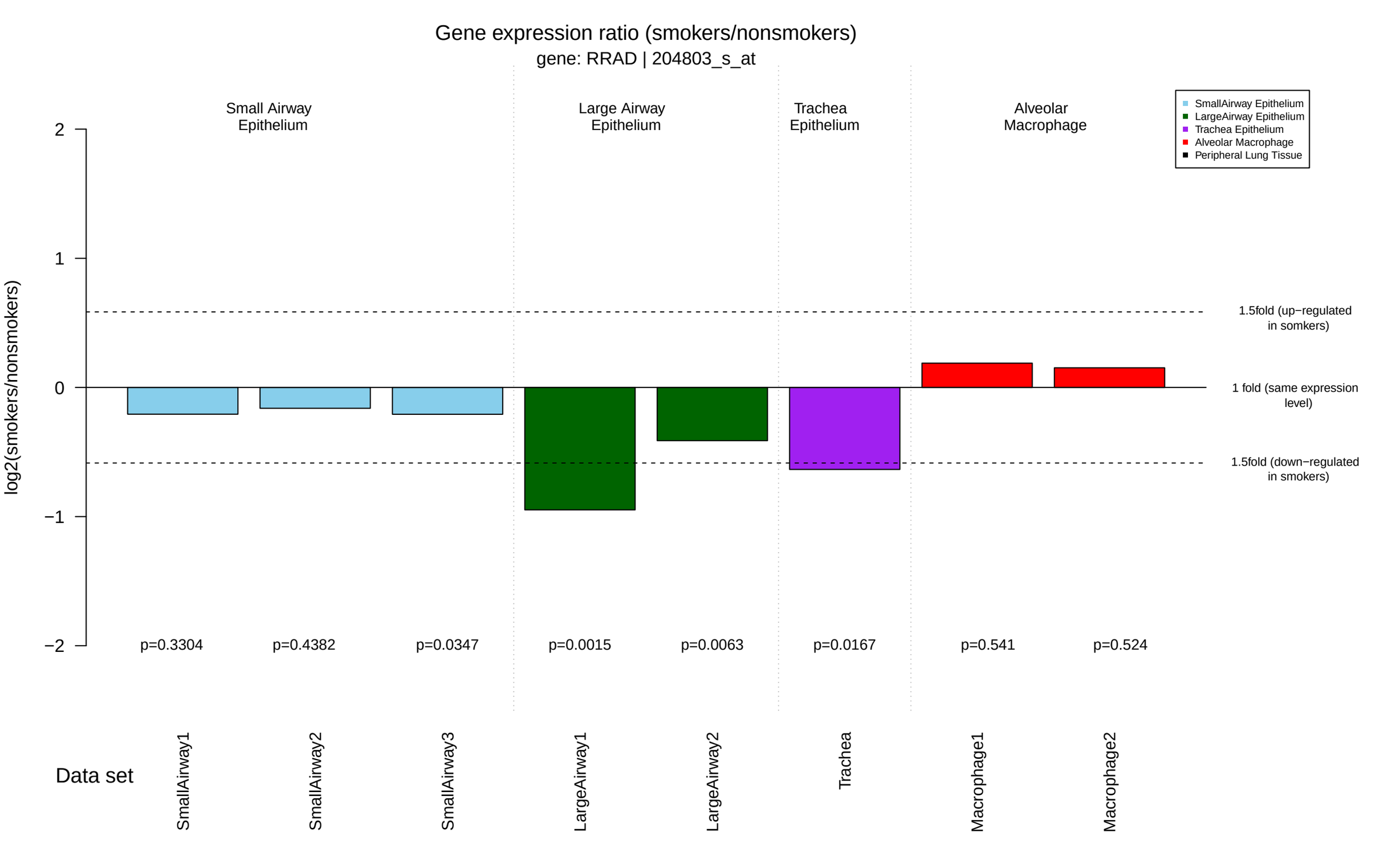
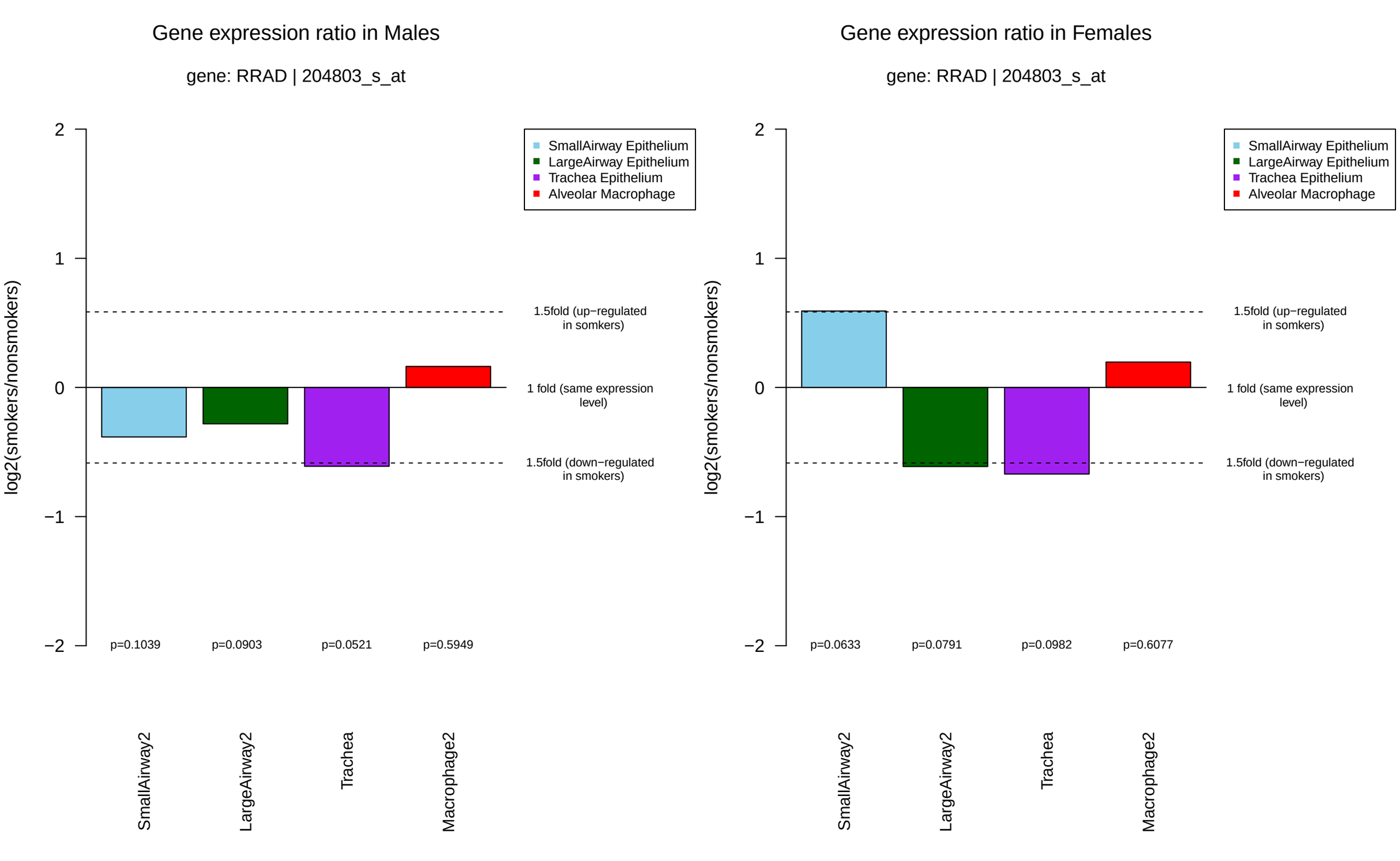
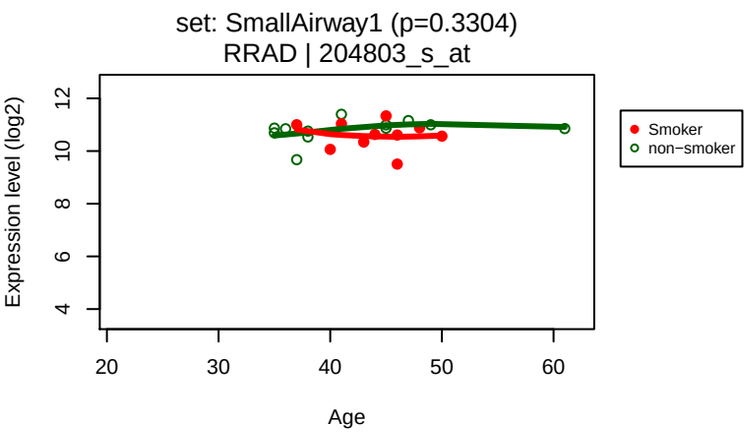
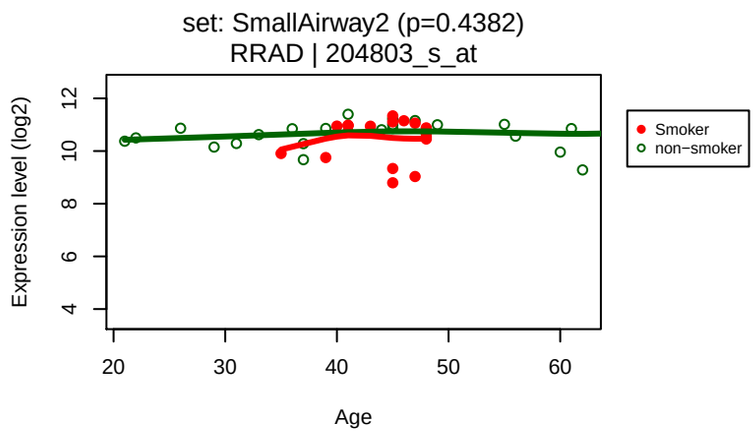
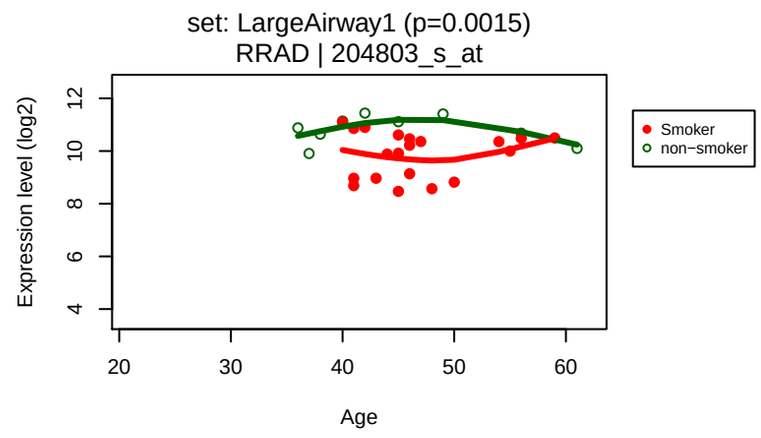

Gene expression of smokers and nonsmokers
gene: RRAD | 204803_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3304
p=0.4382
p=0.0347
p=0.0015
p=0.0063
p=0.0167
p=0.541
p=0.524
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

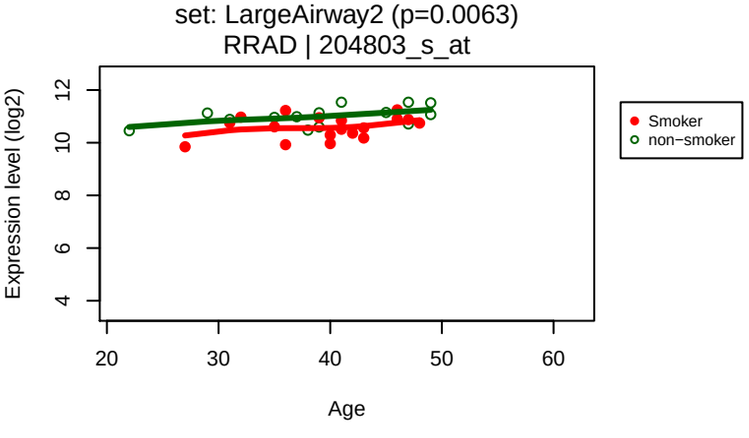

Gene expression ratio (smokers/nonsmokers)
gene: RRAD | 204803_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3304
p=0.4382
p=0.0347
p=0.0015
p=0.0063
p=0.0167
p=0.541
p=0.524
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

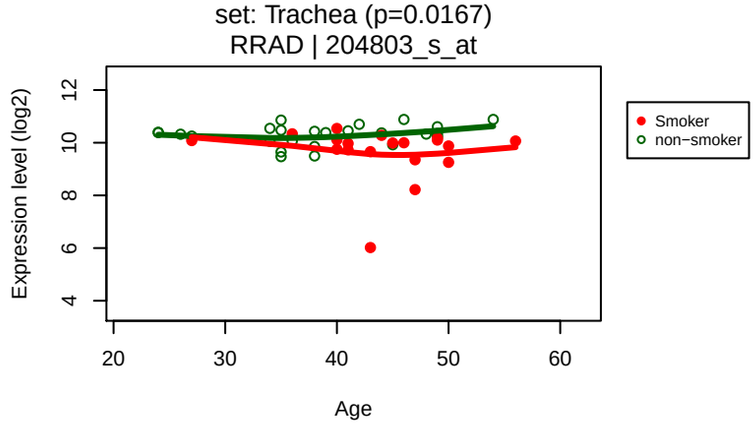

Gene expression ratio in Males
gene: RRAD | 204803_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1039
p=0.0903
p=0.0521
p=0.5949
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRAD | 204803_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0633
p=0.0791
p=0.0982
p=0.6077
SmallAirway2
LargeAirway2
Trachea
Macrophage2

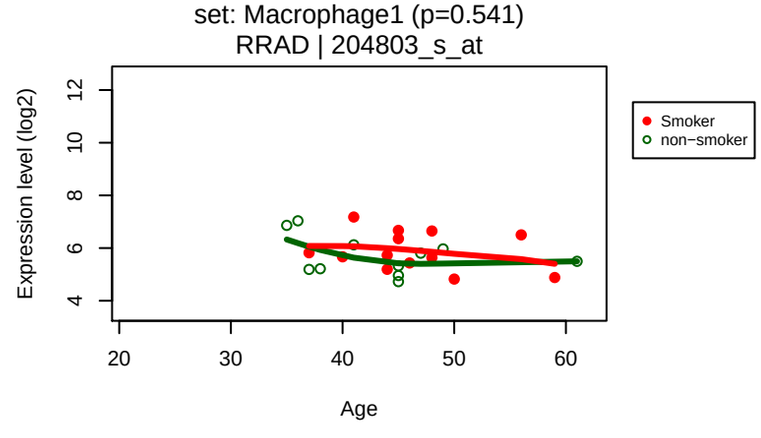

set: SmallAirway1 (p=0.3304)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker

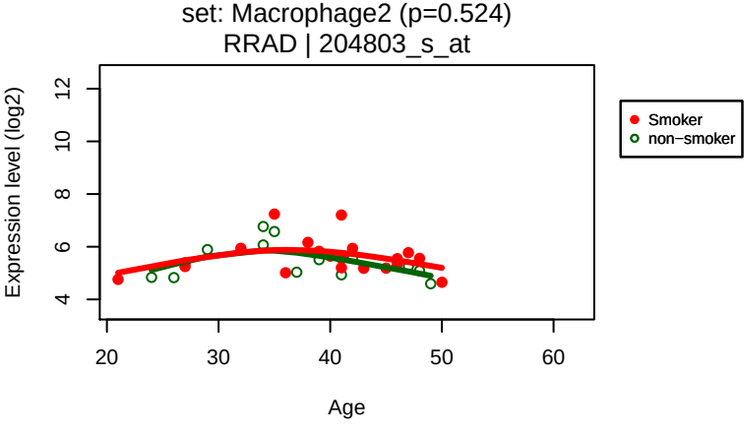

set: SmallAirway2 (p=0.4382)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker

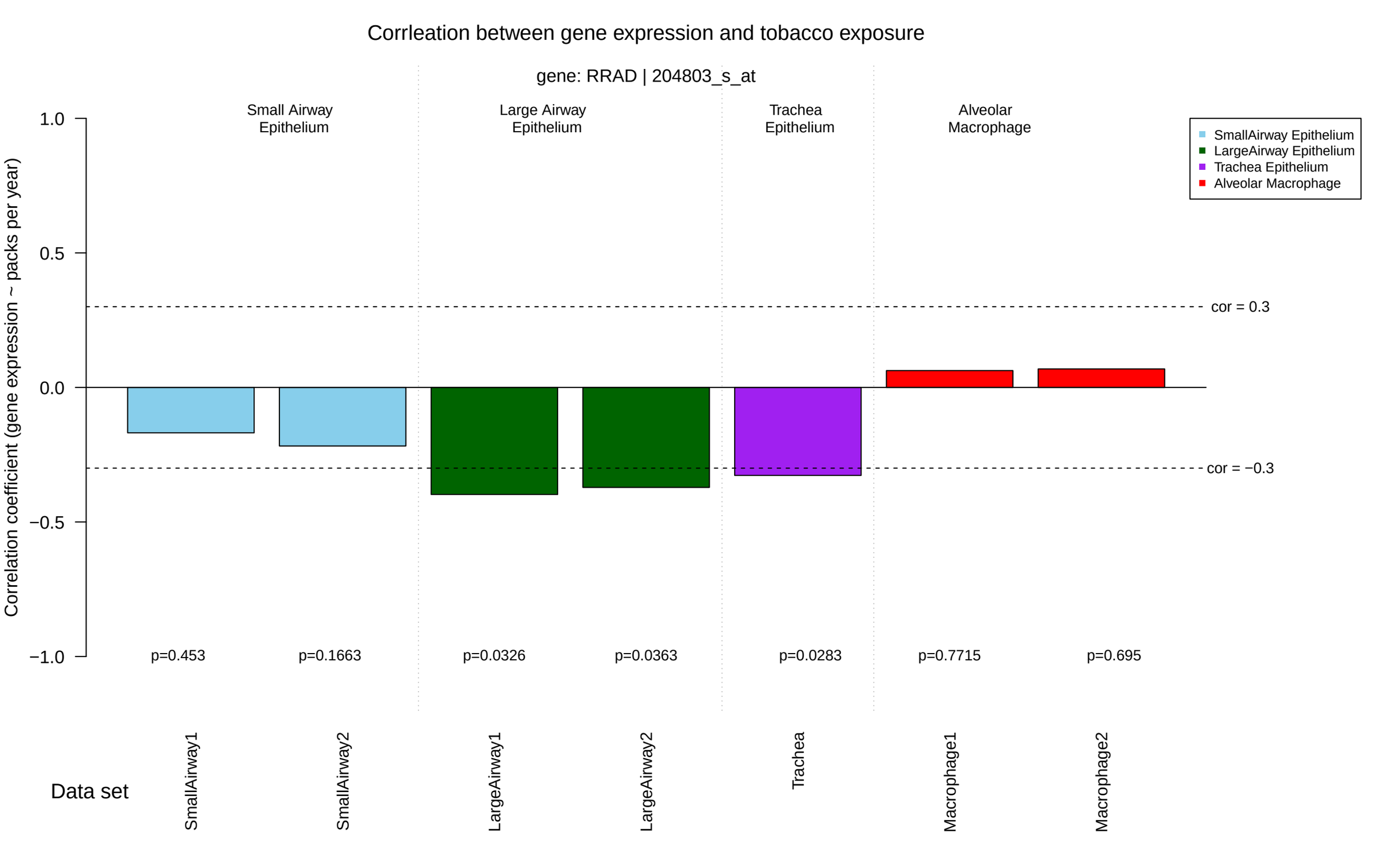

set: LargeAirway1 (p=0.0015)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0063)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0167)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.541)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.524)
RRAD | 204803_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRAD | 204803_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.453
p=0.1663
p=0.0326
p=0.0363
p=0.0283
p=0.7715
p=0.695
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2